

Proposal

Project: Smart Waste Management Portal

Date: October 2, 2026

1. Introduction & Goals

Hi Team,

Thank you for considering DigitalYarr for your digital experience transformation. We're excited to collaborate with you.

A. Introduction

This proposal outlines the development of a customized **Waste Management & Tracking System** designed to digitize and streamline the tracking of municipal waste from collection through its final destination. The objective is to provide a centralized, user-friendly, and traceable platform for managing waste batches across Collection, Transportation, RTS/Transfer, and Processing/Recovery/Disposal stages.

The system will assign a unique Batch ID to each waste batch, allowing its complete journey to be tracked through a centralized timeline. Stage-wise users will record relevant information at each stage, while Admin users will have the ability to manage and correct data when required. Head Officers will have view-only access to monitor the complete waste journey and batch history.

B. Goals

- Build a robust digital platform for creating and tracking waste batches from collection to final destination.
- Implement unique Batch ID-based tracking to maintain a continuous identity for each waste batch throughout its journey.
- Enable stage-wise data entry for Collection, Transportation, RTS/Transfer, and Processing/Final Destination.
- Maintain a complete historical timeline of each waste batch, including relevant locations, quantities, dates, times, and transportation details.
- Implement role-based access control for Collection, Transportation, RTS, Processing/Destination, Admin, and Head Officer users.
- Provide mechanisms for data correction while preserving previous entries and history.
- Offer dashboard views for monitoring total batches, collected, in-transit, RTS, and completed statuses.
- Provide Batch ID search to quickly access the complete journey of a waste batch.
- Enable predefined management of routes, vehicles, and RTS locations for consistent data entry.
- Provide CSV/Excel export for project documentation and analysis.
- Ensure the system is mobile-responsive, allowing field users to enter information from phones as well as laptops.
- Ensure waste tracking data is stored in a structured, reliable, and secure manner.

2. Business Benefit & ROI Analysis

A. Current Challenges:

- Lack of a centralized system for tracking waste from collection to final destination.
- Dependence on manual data entry across different waste-management stages, increasing the possibility of incorrect or incomplete information.
- No centralized view of the complete journey and current status of each waste batch.
- Difficulty maintaining a consistent history when information is added or corrected at different stages.
- Limited role-based control over who can submit, update, or monitor waste-tracking information.
- Manual coordination between Collection, Transportation, RTS, and Processing/Destination stages.
- Limited visibility for management officers to monitor the complete waste journey and batch history.

B. Key Business Impact after digital platform usage:

- Improved operational visibility across the complete waste-management journey.
- Reduced manual coordination and administrative effort through centralized digital data entry.
- Better data accuracy and traceability through structured, stage-wise data collection.
- Complete batch-level history from Collection → Transportation → RTS/Transfer → Processing/Final Destination.
- Improved accountability through role-based access for Collection, Transportation, RTS, Processing, Admin, and Head Officer users.
- Faster identification of incorrect or missing information through centralized monitoring and Admin correction capabilities.
- Improved management visibility through dashboard-based monitoring of batch statuses.
- Easier reporting and analysis through CSV/Excel data export.
- Improved field usability through a mobile-responsive platform for users entering information during collection and transportation activities.00

3. Scope of Work

A. User & Access Management

The system will provide role-based access for the following users:

- **Collection User** - Add collection-stage information for waste batches.
- **Transportation User** - Add transportation, departure, arrival, vehicle, and journey details.
- **RTS User** - Record waste receipt and transfer details at the RTS.
- **Processing / Destination User** - Record processing, recovery, disposal, and final-destination information.
- **Admin** - Full system access, including viewing, managing, and correcting records.
- **Head Officer** - View-only access to monitor complete batch journeys and history.

B. Waste Batch Management

The system will allow the creation and management of individual waste batches.

Each batch will have a unique Batch ID, which will be used to identify and track the complete waste journey.

The batch creation information will include:

- Batch ID
- Date
- Waste Type — Wet / Dry
- Quantity
- Source / Area
- Route
- Vehicle
- System-generated date and time

The system will maintain the batch as a single tracking record throughout its different stages.

C. Collection Management

The Collection User will be able to record the initial collection details of a waste batch, including:

- Collection area / location
- Route
- Vehicle
- Waste type
- Quantity
- Collection date
- Exact collection time
- Driver / Supervisor details

Once submitted, the collection information will become part of the batch's tracking history.

D. Transportation Management

The Transportation User will record the movement of the waste batch from the collection point to the RTS or next designated location.

Transportation information will include:

- Starting location
- Destination
- RTS / next location
- Vehicle
- Departure date and time
- Arrival date and time
- Journey duration
- Related batch information

The journey duration will be calculated based on the recorded departure and arrival times.

E. RTS / Transfer Management

The RTS User will record the receipt and transfer of the waste batch at the designated RTS.

The module will include:

- RTS name
- RTS location
- Arrival date
- Arrival time
- Quantity received
- Waste category
- Handover / receiving details
- Next process / destination

The RTS information will be added to the existing batch history rather than replacing previous collection or transportation information.

F. Processing / Recovery / Final Destination

The Processing / Destination User will record what happens to the waste after transportation and RTS transfer.

The system will capture:

- Processing / destination facility
- Process type
 - Composting
 - Recovery
 - Electricity generation
 - Disposal
- Arrival date
- Arrival time
- Quantity
- Final status

Final status may include:

- Processed
- Recovered
- Disposed
- Completed

G. Batch Tracking & Journey Timeline

The system will provide a Batch ID-based tracking facility.

Users with appropriate access will be able to search for a Batch ID and view its complete journey in a timeline format:

Collection → Transportation → RTS / Transfer → Processing / Recovery / Disposal → Final Status

The tracking view will show the information recorded at each stage, including relevant dates, times, locations, quantities, vehicles, and status information.

H. Record History & Corrections

The system will maintain the history of information entered against each batch.

For operational users:

- Submitted records cannot be directly edited or deleted.
- If incorrect information is entered, the user can submit a corrected entry for the same stage.
- The latest corrected entry will be displayed as the current/valid information.
- Previous entries will remain available as part of the history.

For the Admin:

- Admin can review incorrect or missing information.
- Admin can edit/update records when required.
- Administrative corrections will also be reflected in the batch history.

This approach ensures that the original tracking history is not lost when corrections are made.

I. Dashboard & Monitoring

A centralized dashboard will provide an overview of waste batches and their current stages.

The dashboard will provide visibility into information such as:

- Total waste batches
- Collected batches
- Batches in transportation
- Batches at RTS
- Processed / recovered / disposed batches
- Completed batches

The dashboard will allow authorized users and management to understand the overall status of waste movement.

J. Master Data Management

The Admin will be able to manage predefined master data used throughout the system, including:

- Routes
- Vehicles
- RTS locations

- Other required predefined operational values

This will help maintain consistency in data entry across different stages.

K. Reporting & Data Export

The system will provide reporting/export functionality for operational and management purposes.

The system will support:

- Batch-wise information
- Tracking history
- Stage-wise records
- Current batch status
- CSV / Excel export

The exported data can be used for further analysis and administrative reporting.

L. Responsive Web Application

The system will be developed as a responsive web application that can be accessed from:

- Desktop / Laptop
- Tablet
- Mobile devices

The data-entry screens will be designed to support field users who may access the system from mobile devices.

M. Overall Workflow

The complete system workflow will be:

Create Waste Batch

↓

Collection

↓

Transportation

↓

RTS / Transfer*

↓

Processing / Recovery / Disposal

↓

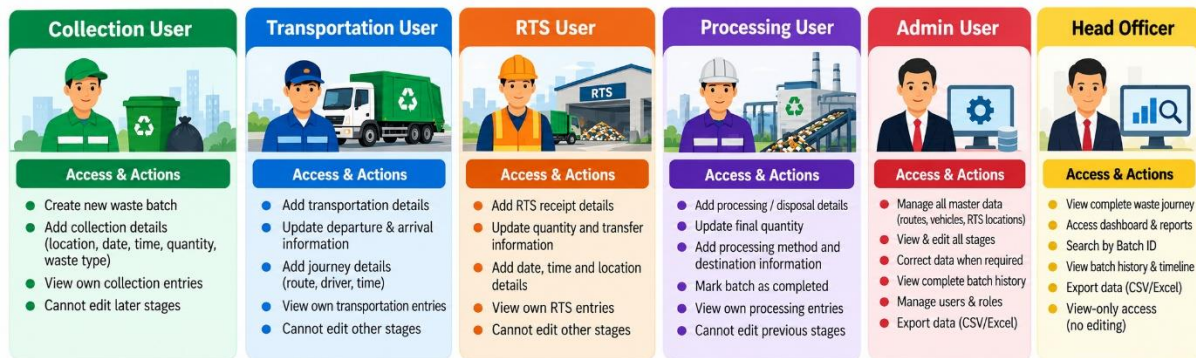
Final Status

↓

Batch History & Reporting

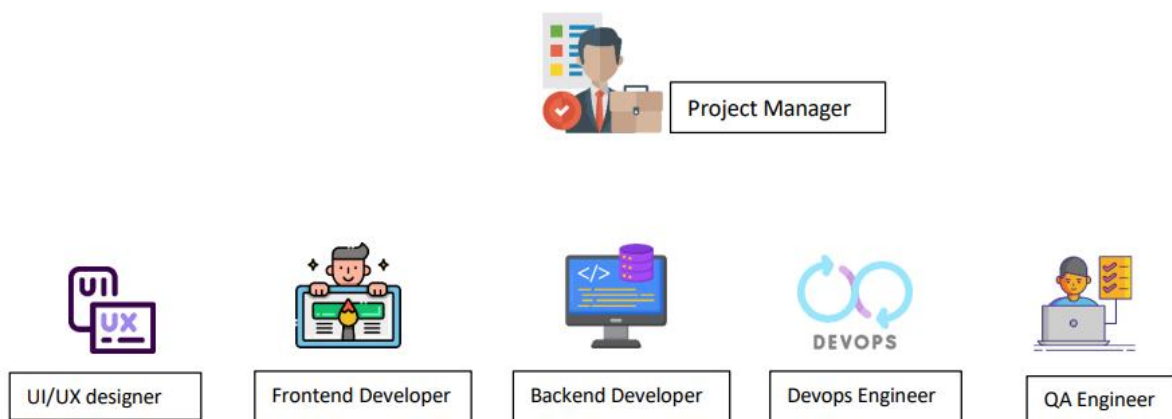
The primary objective of the system is to ensure that **the complete journey and history of each waste batch can be viewed in one centralized platform using its Batch ID**.

4. End-to-End Waste Management Workflow



5. Project Team Structure

Our experienced team follows an agile methodology to move your idea to reality:



- **Project Manager:** Oversees planning, timelines, and delivery.
- **Frontend Developer:** Builds the UI and ensures responsive design.
- **Backend Developer:** Develops logic, APIs, and database integration.
- **UI/UX Designer:** Crafts intuitive screens for consumers and admins.
- **QA/Test Engineer:** Ensures functionality, performance, and security.
- **DevOps Engineer:** Handles deployment and performance monitoring.

6. Budget

Estimated development cost as per current requirements:

Category	DigitalYarr Special Price
Frontend And UI	₹15,000
Backend and Calculation logic and Role based Access Management	₹35,000
Testing and Quality Assessment	₹20,000
Total	₹70,000

**** Special Note:** Negotiations are welcomed.

Payment Milestones

Milestone	Payment Stage	Amount
Milestone 1	Project confirmation & development kickoff - Advance payment	₹5,000
Milestone 2	After Finalizing the UI	₹5,000
Milestone 3	Frontend & UI development completed	₹15,000
Milestone 4	Backend & Calculation Logic completed	₹20,000
Milestone 5	Role-Based Access Management completed	₹15,000
Milestone 6	Testing, Quality Assessment	₹10,000
Total	After Payment We handover the final code.	₹70,000

Payment Terms

- ₹5,000 advance payment is required to initiate the project and reserve the development resources.
- **Due to the short project delivery timeline and priority development requirement, the pricing includes additional effort for accelerated development, testing, and delivery.**
- The remaining payment will be collected upon completion of each respective milestone.
- The above pricing is based on the currently agreed scope of work and requirements discussed for the project.
- Any new features, additional modules, changes in the approved workflow, or requirements outside the agreed Scope of Work will be considered additional work and will be charged separately based on the effort required.
- Significant changes to already completed or approved modules may also result in additional development charges and timeline adjustments.
- Third-party services, paid APIs, domain, production hosting, cloud infrastructure, or other external service charges, if required, will be charged separately unless specifically mentioned in the proposal.
- The project timeline is dependent on timely feedback, approvals, content/data, and access required from the client.
- Final handover will be provided after completion of the agreed scope and settlement of the applicable milestone payments.

7. Special Offerings & Maintenance

As a partner, we provide these complimentary services:

Service	Description
2-Month Maintenance	FREE support for 60 days after launch for smooth functioning.
Training Sessions	Live online training for staff on portal management.
User Manuals	Customized step-by-step guides for consumers and admins.
Free Version Upgrades	Minor enhancements at no cost for the first 60 days.
Basic Security Audit	Initial review to safeguard your application data.

8. Next Steps

1. **Confirm:** Reply to this proposal with your chosen option to receive the formal agreement.
2. **Onboard:** Complete the onboarding form and process the 20% advance payment.
3. **Kick-off:** We will conduct a kick-off call to align on priorities and timelines.